

Describing the Cleaned NHANES Dataset

Anita Adjei

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1 Introduction

The National Health and Nutrition Examination Survey (NHANES) is a programme of studies run by the US National Center for Health Statistics that combines interviews and physical examinations to assess the health and nutritional status of the US population (Centers for Disease Control and Prevention, National Center for Health Statistics 2026). This report describes the cleaned NHANES extract used in this course, which covers 29,400 participants across three waves (2015, 2016, 2017). It summarises who is in each wave (sample size, sex and ethnicity) and how average systolic blood pressure compares across waves.

The dataset keeps the original NHANES labels `gender` and `ethnicity`. These actually capture biological sex and race, so I refer to them as sex and ethnicity (as labelled) below.

2 Participants in each wave

Table 1 gives the number of participants in each wave, the number and percentage of males, and the number and percentage of participants in each ethnic group. Percentages are calculated within each wave.

Table 1: Sample size, sex and ethnicity of NHANES participants by wave

	2015		2016		2017	
	n	%	n	%	n	%
Sample size						
All participants	10,175		9,971		9,254	
Sex						
Male	5,003	49.2	4,892	49.1	4,557	49.2
Ethnicity						
Mexican	1,730	17.0	1,921	19.3	1,367	14.8
Other Hispanic	960	9.4	1,308	13.1	820	8.9
White	3,674	36.1	3,066	30.7	3,150	34.0
Black	2,267	22.3	2,129	21.4	2,115	22.9
Asian	1,074	10.6	1,042	10.5	1,168	12.6
Other	470	4.6	505	5.1	634	6.9

Note: Percentages are within wave. Ethnicity uses the six-category NHANES variable (RIDRETH3).

Each wave contains between 9,254 and 10,175 participants. The sex balance is stable, with males making up 49.1% to 49.2% of each wave. White participants are the largest ethnic group in every wave (30.7% to 36.1%), followed by Black and Mexican participants, so the three waves have a similar composition.

3 Average systolic blood pressure by wave

Each participant’s average systolic blood pressure is the mean of up to four readings, and is missing only if no reading was recorded. Table 2 summarises it for each wave using only participants with a value.

Table 2: Average systolic blood pressure (mm Hg) by wave

Wave	N	Minimum	Maximum	Median	Mean	SD
2,015	7,528	64.7	228.7	115.3	118.3	18.1
2,016	7,363	74.0	231.3	117.3	120.4	18.4
2,017	6,713	72.7	234.0	118.0	121.7	20.3

Note: Mean is highlighted. SD = standard deviation. N = participants with at least one blood pressure reading.

Mean systolic blood pressure is highest in wave 2017 (121.7 mm Hg) and lowest in wave 2015 (118.3 mm Hg), a difference of 3.3 mm Hg. In every wave the median is a little below the mean, which suggests a right-skewed distribution with a tail of high readings (maximum above 220 mm Hg in all waves). The standard deviation is about 18 to 20 mm Hg in each wave, so variability is similar across waves.

4 Summary

The three NHANES waves are similar in size, sex balance and ethnic composition, so differences in blood pressure between waves are unlikely to be due to large shifts in who was sampled. These tables are descriptive and do not account for age, which differs between people and strongly affects blood pressure.

References

Centers for Disease Control and Prevention, National Center for Health Statistics. 2026. *National Health and Nutrition Examination Survey (NHANES)*. <https://www.cdc.gov/nchs/nhanes/>.